

MoonTrace: Structured Logging and Span Tracing for MoonBit

OSC2026 Proposal

Background

MoonBit programs need a small observability layer for tests, CLI tools, services, and WebAssembly apps. MoonTrace provides structured logs, trace context, span lifecycle APIs, and portable export formats.

Goals

- Stable log levels: Trace, Debug, Info, Warn, Error.
- Key/value fields for machine-readable records.
- Context propagation through `trace_id` and `span_id`.
- Tracer, Span, and FinishedSpan APIs for nested work.
- JSONL export and Chrome Trace timeline export.

Technical Plan

The core package remains zero-dependency and portable across MoonBit targets. The proposal-stage code keeps deterministic timestamps for tests; later milestones add pluggable clocks, JSON escaping, nested span parenting, and richer exporters. Explicit data types are preferred over global state.

Milestones

- 2026-07-10: publish repository, submit proposal, keep a minimal compiling demo.
- 2026-07-15: apply review feedback and freeze the v0.1 API shape.
- 2026-08-15: complete logger, context, span nesting, JSONL, Chrome Trace, and tests.
- 2026-09-20: publish to Mooncakes, add examples, final report, and demo material.

Acceptance Criteria

moon check and moon test pass in CI. README examples match implementation. JSONL output is valid JSON per line. Chrome Trace output opens in browser timeline tooling. Mooncakes package contains docs, examples, license, and changelog.

Risk Control

Keep the public API small, rely on official moon.mod.json/moon.pkg formats, add snapshot tests for exporters, and emphasize trace context plus timeline visualization beyond a generic logger.